

AWS Backup – Remove Deleted EC2 Instance from Backup Resource Assignment

1. Purpose

This document explains how to remove a **deleted/terminated EC2 instance** from an AWS Backup resource assignment when AWS Backup is generating failed backup jobs for that instance.

Example scenario

An EC2 instance is deleted:

Instance ID: i-0fe8b9bda4ae9f6e3

But the instance is still included in the AWS Backup resource assignment.

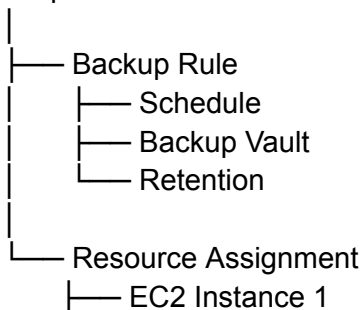
AWS Backup then tries to back up the deleted instance and the backup job fails with:

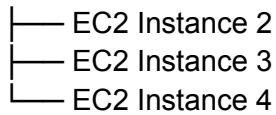
The instance i-0fe8b9bda4ae9f6e3 is not available or does not exist

2. AWS Backup Structure

Understand the relationship first:

Backup Plan





Simple explanation

- **Backup Plan** → Overall backup configuration.
- **Backup Rule** → When and how long backups are retained.
- **Backup Vault** → Where backups are stored.
- **Resource Assignment** → Which EC2 instances should be backed up.

For this activity, we only need to modify the **Resource Assignment**.

3. Example From the Actual Activity

AWS Account

433957317038

Region

ap-south-1
Mumbai

Backup Plan

Prod-Backup-Schedule-3AM

Backup Rule

daily-3am

Backup Vault

Prod-Backupvault

Schedule

03:00 AM
Asia/Kolkata
Daily

Retention

2 days

4. Failed Backup Details

The failed backup job contained:

Backup Job ID:
ADC4DB89-3F37-5F64-06F7-89B772CE08CB

Status:
FAILED

Resource Type:
EC2

Instance:
i-0fe8b9bda4ae9f6e3

Failure message:

The instance i-0fe8b9bda4ae9f6e3 is not available or does not exist

This indicated that AWS Backup was still trying to back up an EC2 instance that had already been deleted.

5. Step 1 – Verify the EC2 Instance

Go to:

AWS Console

→ EC2
→ Instances

Search for the Instance ID:

i-0fe8b9bda4ae9f6e3

Confirm that the instance has actually been terminated/deleted.

Do not remove an instance from backup just because of a temporary backup failure.

6. Step 2 – Open the Backup Plan

Go to:

AWS Console
→ AWS Backup
→ Backup plans

Open:

Prod-Backup-Schedule-3AM

Verify that the existing backup rule is correct:

Rule:
daily-3am

Schedule:
03:00 AM Asia/Kolkata

Backup Vault:
Prod-Backupvault

Retention:
2 days

Do not change the backup rule, schedule, vault, or retention as part of this cleanup.

7. Step 3 – Open Resource Assignments

Inside the backup plan, go to:

Resource assignments

Open the existing assignment:

Prod-Backup-Schedule-3AM

The old assignment contained:

i-00c4b3f98828903e
i-023be3156e3e5df6a
i-0411cec8d11b0b233
i-06e53580bd1703227
i-08cc8aad04d1955a
i-0fe8b9bda4ae9f6e3 ← Deleted instance

8. Important – Do Not Delete the Old Assignment Immediately

In this situation, the existing resource assignment could not be edited to remove only one EC2 instance.

The available option was to delete the complete resource assignment.

Therefore, deleting the old assignment first could temporarily remove all six resources from that assignment.

Safe approach

Create NEW assignment

↓

Add only required EC2 instances

↓

Verify NEW assignment



Delete OLD assignment

9. Step 4 – Create a New Resource Assignment

Inside the same Backup Plan, create a new resource assignment.

Example name:

Prod-Backup-Schedule-3AM-Updated

Use the same backup IAM role.

Select only the required active EC2 instances:

i-00c4b3f98828903e
i-023be3156e3e5df6a
i-0411cec8d11b0b233
i-06e53580bd1703227
i-08cc8aad04d1955a3

Do not select:

i-0fe8b9bda4ae9f6e3

because this instance has already been deleted.

10. Step 5 – Verify the New Assignment

Open:

Prod-Backup-Schedule-3AM-Updated

Confirm that the five required instances are present:

- ✓ i-00c4b3f98828903e
- ✓ i-023be3156e3e5df6a
- ✓ i-0411cec8d11b0b233
- ✓ i-06e53580bd1703227
- ✓ i-08cc8aad04d1955a3

Confirm that the deleted instance is NOT present:

- ✗ i-0fe8b9bda4ae9f6e3

This is the most important verification before deleting the old assignment.

11. Step 6 – Final Safety Check

Before deleting the old assignment, verify:

Account

433957317038

Region

Mumbai / ap-south-1

Backup Plan

Prod-Backup-Schedule-3AM

New Assignment

Prod-Backup-Schedule-3AM-Updated

New Assignment

5 required instances ✓
Deleted instance absent ✓

Make sure you are going to delete:

Prod-Backup-Schedule-3AM

and NOT:

Prod-Backup-Schedule-3AM-Updated

12. Step 7 – Delete the Old Assignment

Go to:

AWS Backup

→ Backup Plans

→ Prod-Backup-Schedule-3AM

→ Resource assignments

Select:

Prod-Backup-Schedule-3AM

Click:

Delete

Confirm the deletion.

Important

Delete only the **old resource assignment**.

Do NOT delete:

Backup Plan

Backup Rule

Backup Vault

Recovery Points

13. Step 8 – Verify After Deletion

Refresh the Resource Assignments page.

Expected result:

Prod-Backup-Schedule-3AM-Updated

should remain.

The old assignment:

Prod-Backup-Schedule-3AM

should no longer be present.

14. What Happens After This?

The new resource assignment is attached to the same backup plan/rule.

Therefore, the next scheduled backup will use the new assignment and select the five remaining EC2 instances.

The deleted instance:

i-0fe8b9bda4ae9f6e3

is no longer part of the backup selection.

Creating the new assignment does not immediately start a backup. It only changes which resources are selected for future scheduled backups.

15. Existing Recovery Points

Deleting the resource assignment does not automatically mean existing recovery points are deleted.

There is a difference between:

Resource Assignment



Which resources AWS Backup should back up

and:

Recovery Point



The actual stored backup

Therefore, changing/deleting a resource assignment is not the same as deleting the existing backups.

16. Complete Process – Quick Reference

Use this checklist for future tickets:

- ☐ Check failed AWS Backup job
- ☐ Identify failed EC2 Instance ID
- ☐ Verify the EC2 instance is actually deleted
- ☐ Open the correct AWS Backup plan
- ☐ Check the existing backup rule
- ☐ Open Resource Assignments
- ☐ Identify the deleted instance in the old assignment
- ☐ Create a NEW resource assignment
- ☐ Add only the required active instances
- ☐ Confirm deleted instance is NOT included
- ☐ Verify the new assignment
- ☐ Delete the OLD resource assignment
- ☐ Refresh and verify the NEW assignment remains
- ☐ Do not change Backup Plan/Rule/Vault/Retention

17. Simple Example for Future Use

Suppose the backup assignment contains:

Production-Backup

```
|  
├── i-11111111  
├── i-22222222 ← EC2 deleted  
├── i-33333333  
└── i-44444444
```

Backup fails because:

i-22222222 does not exist

Create a new assignment:

Production-Backup-Updated

```
|  
├── i-11111111  
├── i-33333333  
└── i-44444444
```

Verify it.

Then delete:

Production-Backup

Final result:

Production-Backup-Updated

```
|  
├── i-11111111  
├── i-33333333  
└── i-44444444
```

The deleted instance is no longer selected.

18. Key Rule to Remember

When an EC2 instance is deleted but is still present in an AWS Backup resource assignment, create a replacement assignment containing only the active resources, verify it, and then delete the old assignment.

This prevents accidentally removing the backup configuration or losing the correct resource selection.